Polynomial Functions and Equations

Concept

Zeros, multiplicities and the leading term explain the main shape of a polynomial graph.

End behaviour

Degree parity and leading-coefficient sign control the two ends.

Multiplicity

Odd zero: crosses; even zero: touches the x-axis.

Factor / remainder

Dividing by $x - a$ leaves remainder $P(a)$; $P(a) = 0$ means a factor.

Division

Include zero coefficients for missing powers in synthetic division.

Example 1

Factor $P(x) = x^3 - 4x^2 + x + 6$ and solve $P(x) = 0$.

Solution

1. Test $x = 2$: $P(2) = 0$, so $x - 2$ is a factor.
2. Synthetic division: coefficients $1, -4, 1, 6$ with 2 give $1, -2, -3$ and remainder 0.
3. Quotient $x^2 - 2x - 3 = (x - 3)(x + 1)$.
4. $P(x) = (x - 2)(x - 3)(x + 1)$; roots $-1, 2, 3$.

Polynomial Functions and Equations

Example 2

Describe $y = -(x + 1)^2(x - 2)$.

Solution

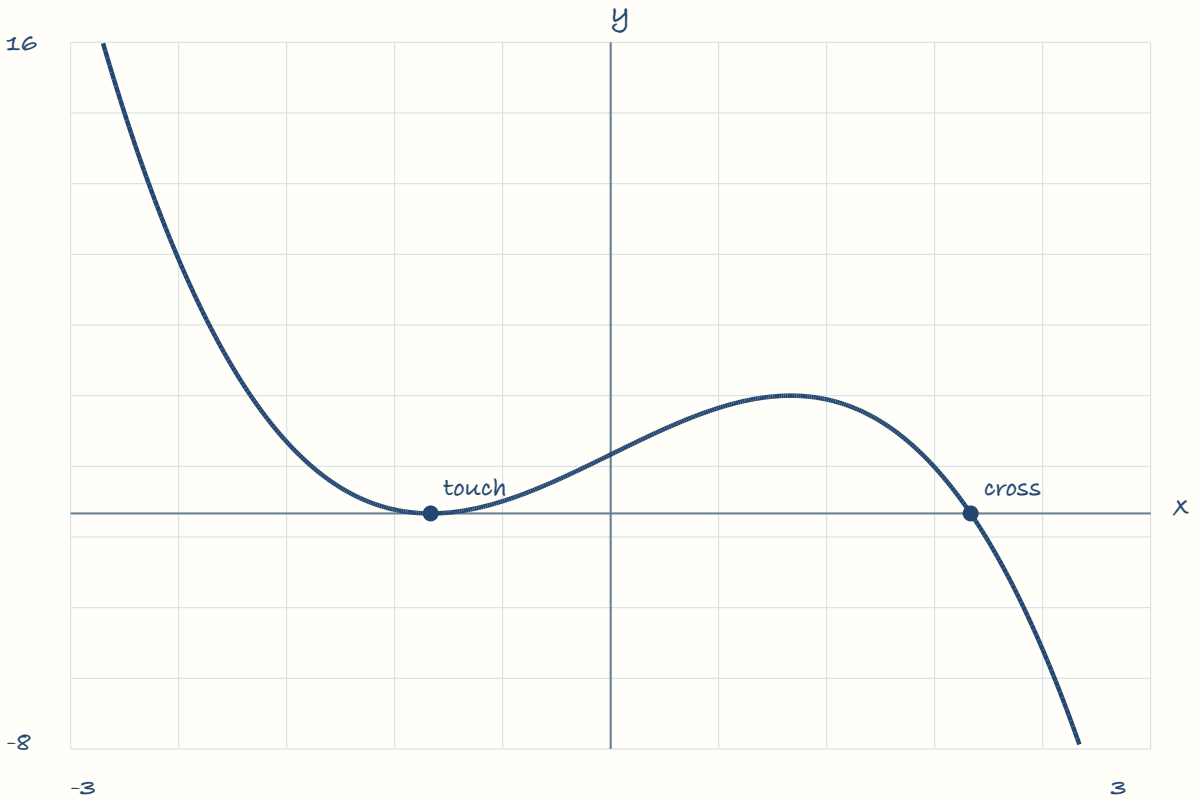
1. Degree 3, negative leading coefficient: left end up, right end down.

2. Touch at $x = -1$; cross at $x = 2$. y-intercept $= 2$.

Polynomial Functions and Equations

Visual notes | read the graph

$$y = -(x + 1)^2(x - 2)$$



The double zero touches. The simple zero crosses. The leading negative cubic term controls the ends.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Polynomial Functions and Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find the remainder of $x^3 + 2x^2 - 5$ at division by $x - 1$.

2. Describe ends and zero behaviour of $y = 2(x - 1)^2(x + 3)^2$.

3. Factor $x^3 - 3x^2 - 4x + 12$.

4. Solve $x^3 = 4x$.

Polynomial Functions and Equations

Practice answers | check after trying

1. $P(1) = 1 + 2 - 5 = -2$.

2. Degree 4, leading coefficient positive: both ends up. It touches at -3 and 1.

3. Grouping gives $x^2(x - 3) - 4(x - 3) = (x - 3)(x - 2)(x + 2)$.

4. $x(x^2 - 4) = x(x - 2)(x + 2) = 0$, so $-2, 0, 2$. Dividing by x without checking would lose the zero solution.